COMPUTER NETWORK

SDN-PROJECT

PROJECT TITLE

Network Delay Measurement Tool using SDN:

AIM

To measure the delay of packets in a network using Software Defined Networking (SDN)

1. POX Controller:

```
pavakumarh@PavankumarH:~$ cd ~/pox
pavakumarh@PavankumarH:~/pox$ ./pox.py forwarding.l2_learning misc.delay_measure
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
INFO:misc.delay_measure:Delay Measurement Module Loaded
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 2] connected
INFO:misc.delay_measure:Delay for switch 1: 154.656172 ms
INFO:misc.delay_measure:Delay for switch 1: 35.414219 ms
INFO:misc.delay_measure:Delay for switch 1: 828.523636 ms
INFO:misc.delay_measure:Delay for switch 1: 0.830412 ms
INFO:misc.delay_measure:Delay for switch 1: 191.346407 ms
INFO:misc.delay_measure:Delay for switch 1: 0.568867 ms
INFO:misc.delay_measure:Delay for switch 1: 606.189251 ms
INFO:misc.delay_measure:Delay for switch 1: 385.369539 ms
INFO:misc.delay_measure:Delay for switch 1: 2883.086205 ms
INFO:misc.delay_measure:Delay for switch 1: 3.201485 ms
INFO:misc.delay_measure:Delay for switch 1: 5.108833 ms
INFO:misc.delay_measure:Delay for switch 1: 3.039598 ms
INFO:misc.delay_measure:Delay for switch 1: 177.451849 ms
INFO:misc.delay_measure:Delay for switch 1: 256.190300 ms
INFO:misc.delay_measure:Delay for switch 1: 4607.617378 ms
INFO:misc.delay_measure:Delay for switch 1: 1.920223 ms
INFO:misc.delay_measure:Delay for switch 1: 3325.647593 ms
INFO:misc.delay_measure:Delay for switch 1: 768.163919 ms
^CINFO:core:Going down...
INFO:openflow.of_01:[00-00-00-00-00-01 2] disconnected
INFO:core:Down.
pavakumarh@PavankumarH:~/pox$
```

2. Start Mininet

```
PS C:\Users\pavan> wsl
pavakumarh@PavankumarH:/mnt/c/Users/pavan$ sudo mn --topo single,2 --controller=remote,ip=127.0.0.1
[sudo] password for pavakumarh:
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=14.6 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.320 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.107 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.116 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.098 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.134 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.098 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.126 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.121 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.109 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.147 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.088 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.128 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.072 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.128 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.187 ms
^C
--- 10.0.0.2 ping statistics ---
22 packets transmitted, 22 received, 0% packet loss, time 21469ms
rtt min/avg/max/mdev = 0.064/0.776/14.617/3.020 ms
mininet>
```